**S4 Table - Part A. Complete standardized form.** Original publications of the tools externally validated in 25 articles.

| Score | First publication | Name of the cohort Country (nb of centres) | Date of inclusion | Initial type of injury | Initial main outcome | Type of tool | Predictors | Public equation/ score? | External validation |
|---|---|---|---|---|---|---|---|---|---|
| Tuhrim equation | Tuhrim et al. 1991 Ann Neurol | Pilot Stroke Data Bank USA (4) | NA | ICH 100% | - Mortality at 1 month - Mortality or Bartel index > 60 at 1 year | Continuous score | 4 Pulse pressure, GCS, ICH volume, IVH, IVH*GCS | Yes | Edwards et al. 1999 Neurology (1) |
| ICH score | Hemphill et al. 2001 Stroke | San Francisco USA (2) | 1997 - 1998 | ICH 100% | Mortality at 1 month | Risk stratification scale based on the strength of association with outcome from LR model | 5 Age (2 cat), GCS (3 cat), ICH volume (2 cat), IVH (y/n), Infra-tentorial origin (y/n) | Yes | Barbieri et al. 2009 J Eval Clin Pract (2)<br>Chuang et al. 2009 Int J Qual Health Care (3)<br>Di Napoli et al. 2011 Stroke (4)<br>Godoy et al. 2006 Stroke (5)<br>Huang et al. 2012 Eur J Neurol (6)<br>Maas et al. 2017 Cerebrovasc Dis (7)<br>Naval et al. 2009 Neurol Res (8)<br>Patriota et al. 2009 Arq Neuropsiquiatr (9)<br>Rodriguez-Fernandez et al 2018 BMJ open (10)<br>Schmidt et al 2018 Neurology (11)<br>Sembill et al 2017 Neurology (12) |
| ICH-GS | Ruiz-Sandoval et al. 2007 Stroke | Mexico (1) | 1999 - 2003 | ICH 100% | - Mortality at discharge - Mortality at 1 month | Score (5-13) | 5 age (3 cat), GCS (3 cat), ICH location (2 cat), IVH volume (3 | Yes | Chuang et al. 2009 Int J Qual Health Care (3) |

| Score | First publication | Name of the cohort Country (nb of centres) | Date of inclusion | Initial type of injury | Initial main outcome | Type of tool | Predictors | Public equation/ score? | External validation |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | - Fav outcome (GOS 4-5) at 1 month | | cat), extension into ventricles (y/n) | | Naval et al. 2009 Neurol Res (8) |
| Modified Intracerebral Hemorrhage Score (MICH) | Cho et al. 2008 Crit Care Med | Taiwan (1) | 2001 - 2005 | ICH 100% | - Mortality at 6 month - Fav. outcome (GOS 4-5) at 12 months - Barthel index (≥ 55) at 12 months | Score (0-5) | 3 GCS (3 cat), ICH volume (3 cat), IVH or hydrocephalus (y/n) | Yes | Sembill et al 2017 Neurology (12) |
| Max ICH score | Sembill et al 2017 Neurology | Germany (monocentric) | 2007/01 - 2011/12 | Maximally treated ICH 100% | - Mortality at 3 and 12 months - Unfav outcome (mRS 4-6) at 3 and 12 months | Continuous score | 6 lobar ICH vol (2 cat), non-lobar ICH vol (2 cat), age (4 cat), NIHSS (4 cat), IVH (y/n), oral anticoag (y/n) | Yes | Schmidt et al. 2018 Neurology (11) |
| ISAT | Risselada et al. 2010 Eur J Epidemiol | ISAT Europe (RCT - multicentre) | NA | aneurysmal SAH 100% | Mortality at 60 days | Continuous score | 4 age (in decades), lumen size (num), Fisher grade (4 cat), and WFNS grade (5 cat + NA) | Yes | Dijkland et al. 2016 Crit Care Med (13) |
| VASOGRADE | de Oliveira Manoel et al. 2015 Stroke | International 3 SAHIT trials (CONSCIOUS-1, EPO trial, statin trial) + 1 centre (Canada) | NA | SAH 100% | Delayed Cerebral Ischemia | 3- cat grading system (green - yellow - red) | 2 WFNS (3 cat), modified Fisher Scale (2 cat) | Yes | Dengler et al. 2017 Eur J Neurol (14) |
| SAH sum score / Hijdra score | Hijdra et al. 1988 Stroke | European (Rotterdam, Amsterdam, Glasgow, London) (Multicenter) | 1977-1983 | aneurysmal SAH 100% | - Unfav outcome (GOS 1-3) at 4 weeks- DCI - rebleeding | Continuous score | 2 SAH volume (num), GCS (num) | Yes | Claassen et al. 2004 Crit Care Med (15) |
| SAHIT | Jaja et al. 2018 BMJ | SAHIT International (7 RCT = Van den Bergh 2005, IMASH, COUNSCIOUS-1, ISAT, IHAST, MAPS, Etminan 2013 + 2 registries HELBOK 2013, Smith 2005, Reilly 2004 - multicenter) | NA | SAH 100% | - Mortality at 3 months - Unfav outcome (GOS 1-3) at 3 months | Logistic regression | - Core 3: age, premorbid history of hypertension, WFNS on adm - Neuroimaging 6: core + CT vol of SAH, aneurysmal size, aneurysm location - Full 7: neuroimaging + treatment modality | Online calculator | Mascitelli et al 2018 Neurosurgey (16) |
| HAIR / SAH score | Lee et al. 2014 Neurocrit Care | Chicago USA (1) | 2006 - 2011 | SAH 100% Exclusion of CT negative | In-hospital mortality | risk stratification scale based on | 4 Hunt and Hess grade (3 cat), age (3 cat), | Yes | Witsch et al. 2016 Ann Neurol (17) |

| Score | First publication | Name of the cohort Country (nb of centres) | Date of inclusion | Initial type of injury | Initial main outcome | Type of tool | Predictors | Public equation/ score? | External validation |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | the strength of association of the predictors with the outcome (0-8) | IVH (y/n), re-bleeding within 24 hours (y/n) | | Dengler et al. 2017 Eur J Neurol (14)<br>Abulhasan et al. 2017 Neurocrit Care (18) |
| APACHE II | Knaus et al. 1985 Crit Care Med | USA (13) | 1979 - 1982 | All admission in ICU | Hospital mortality | Scale (0 - 71) | 14 age, t°, mean BP, HR, RR, O2, art pH, Na, K, creat, Haematocrit, WBC, GCS, chronic health point | Yes | Claassen et al. 2004 Crit Care Med (15)<br>Fallenius et al. 2017 Scand J Trauma Resusc Emerg Med (19)<br>Huang et al. 2012 Eur J Neurol (6)<br>Moon et al. 2015 J Clin Neurosci (20)<br>Rodriguez-Fernandez et al. 2018 BMJ Open (10) |
| SOFA | Vincent et al. 1996 Intensive Care Med | Expert meeting | NA | Admission in ICU for sepsis | Hospital mortality | Scale (6-24) | 6 PaO2/FiO2, platelets, bilirubin, hypotension, GCS, creatinine | Yes | Fallenius et al. 2017 Scand J Trauma Resusc Emerg Med (19)<br>Basile-Filho et al. 2018 Medicine (21) |
| SIRS summary score | Bone et al. 1992 Chest | USA (40) | NA | Admission in ICU for sepsis | Hospital mortality | NA | 4 t°, HR, RR, WBC count | NA | Claassen et al. 2004 Crit Care Med (15) |
| SAPS | Le Gall et al 1984 Crit Care Med | France (8) | NA | Admission in ICU | Mortality at ICU discharge | Scale (0-50) | 14 Age, HR, sBP, t°, RR or MV, urinary output, urea, haematocrit, WBC count, glc, K, Na, HCO3, GCS | Yes | Handschu et al. 2005 J Neurol (22) |
| SAPS II | Le Gall et al 1993 JAMA | International (multicentre) | 1991 - 1992 | Admission in ICU | Mortality at ICU discharge | Scale (0-163) | 17 age, HR, sBP, t°, MV, urinary output, urea, WBC count, K, Na, HCO3-, bilirubin, GCS, type of | Yes | Schuiling et al. 2005 Neurosurgery (23)<br>Handshu et al. 2005 J Neurol (22) |

| Score | First publication | Name of the cohort Country (nb of centres) | Date of inclusion | Initial type of injury | Initial main outcome | Type of tool | Predictors | Public equation/ score? | External validation |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | admission, AIDS, hemato malignancy, metastatic cancer | | Celi et al. 2012 J Pers Med (24) |
| | | | | | | | | | Czorlich et al. 2015 Acta Neurochir (Wien) (25) |
| | | | | | | | | | Moon et al. 2015 J Clin Neurosci (20) |
| | | | | | | | | | Huang et al. 2012 Eur J Neurol (6) |
| | | | | | | | | | Fallenius et al. 2017 Scand J Trauma Resusc Emerg Med (19) |
| | | | | | | | | | Barbieri et al. 2009 J Eval Clin Pract (2) |
| SAPS III | Moreno et al. 2005 Intensive Care Med | SAPS 3 project European multicentric cohort (303) | 2002 | All admission in ICU | Mortality at discharge | Score (5-124) | (20) Age, comorbidities, LOS before ICU, location before ICU, use of vasoactive drugs before ICU, planned ICU adm, reasons for ICU adm, surgical status, anatomical site, acute infection at adm, GCS, bilirubine, temperature, creatinine, HR, WBC, pH, platelets, systolic BP, oxygenation | Yes | Basile-Filho et al. 2018 Medicine (21) |

**S4 Table - Part B. Complete standardized form.** Details of the 25 articles about stand-alone external validation studies.

| Score | External validation | Study design Country (nb of centre) Date of incl. | Type of injuries | Outcomes | Sample size | Description of the population | Reporting of global performances | Reporting of discrimination | Reporting of classification | Reporting of calibration | Recalibration or updating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Tuhrim equation | Edwards et al. 1999 Neurology (1) | Prospective cohort USA (1) 1996-1997 | Supra-tentorial ICH (exclusion SAH) | Mortality at discharge | 81 mortality 21 (26%) | - Age Mean 62.4 SD 16.1 - Female 42% | No | No | correct prediction survivor = 88% death 62% (Cut off NA) | No | No |
| ICH score | Barbieri et al. 2009 J Eval Clin Pract (2) | Registry of ICU Italy (1) period NA | spontaneous intra parenchymal hemorrhage | Mortality at 1 month | 81 mortality 49 (60.5%) | - Age Mean 64.6 SD 13.8 - Female: 36% | No | AUC ROC 0.732 (0.617-0.847) | No | No | No |
| | Chuang et al. 2009 Int J Qual Health Care (3) | Retrospective cohort Taiwan (1) 2006-2007 | spontaneous ICH | Mortality at 1 month | 293 Mortality 40 (14%) | - Age Mean 60.6 SD 16.7 - Female 30% | accuracy: 74,1% | AUC ROC: 0.74 (0.65-0.83) | Cut off NA (highest Youden) Se, Sp, PPV, NPV, LR+, LR- (table 4) | HL GOF test p>0.05 | No |
| | Di Napoli et al. 2011 Stroke (4) | Prospective Argentine multicenter cohort (2) 2005 – 2009 | Spontaneous ICH | Mortality at 1 month | 210 Mortality 63 (30%) | - Age mean 67.3 SD 11.5 - Female 42% | Nagelkerke $R^2$ 56.9 LR $chi^2$ 144.7 | AUC ROC 0.94 | No | HL GOF test P=0.9 | No |
| | Godoy et al. 2006 Stroke (5) | Prospective cohort Argentina (2) 2003-2004 | spontaneous ICH | Mortality at 1 month / Unfav outcome (GOS 1-3) at 6 months | 153 mortality 53 (35%) | - Age Mean 66 SD 12 - Female 37% | No | AUC ROC 1-m mortality 0.882 (0.830 - 0.934) 6-m GOS 0.844 (0.781 - 0.907) | PPV, NPV ICH scores of 1, 2, 3, and 4 were 2.9%, 30.8%, 61.1%, and 88.2% | No | No |
| | Huang et al. 2012 Eur J Neurol (6) | Registry of NICU Chine (1) 2000-2011 | primary pontine hemorrhage | Mortality at 1 month | 75 mortality 31 (41%) | - Age Mean 54.8 SD 12.7 - Female 21% | No | AUC ROC 0.844 (0.757 - 0.931) | Cut off 1.5 (Youden) Se 96.8% Sp 54.5% | HL GOF test p=0.176 Table obs / pred | No |
| | Maas et al. 2017 Cerebrovasc Dis (7) | Prospective NICU cohort US (1) 2010 - 2016 | Spontaneous ICH | Unfav outcome (mRS 4-6) at 3 months | 254 Good outcome 122 (48%) | - Age Mean 64.3 SD 13.6 - Female 50% | Nagelkerke $R^2$ 0.36 –2 log likelihood 270.7 | No | PPV 66.7% NPV 83.2% Diagnostic effectiveness 73% | No | No |
| | Naval et al. 2009 Neurol Res (8) | Registry of NICU Baltimore USA (1) 1999-2006 | Supra-tentorial, spontaneous ICH Excl of prior mRS 2-5 | Mortality at 1 month | 125 mortality 29 (23%) | - Age median 63.5 range 34-90 - Female 42% | No | No | Cut off ≥ 3 (Proba 50/50) PPV: 71% NPV 97.7% Se 93.1% Sp 88.5% | chi square test obs/pred p=0,14 | No |
| | Patriota et al. 2009 Arq Neuropsiquiatr (9) | Prospective cohort admission ICU Brazil (1) 2006 | spontaneous ICH | Mortality at 1 month / Fav outcome (GOS 4-5) at 12 months | 37 1-m mortality 38% 1-y GOS 4-5 : 38% | - Age Mean 67.7 SD 11.2 - Female 51% | No | AUC ROC mortality 0.804 (0.65 - 0.95) GOS 4-5 0.77 (0.60 - 0.89) | mortality cut off: 3 Se 85.7% Sp 65.2% GOS 4-5 cut off ≤2 Se 100% Sp 42% | Histogram obs / pred | No |

| Score | External validation | Study design Country (nb of centre) Date of incl. | Type of injuries | Outcomes | Sample size | Description of the population | Reporting of global performances | Reporting of discrimination | Reporting of classification | Reporting of calibration | Recalibration or updating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Rodriguez-Fernandez et al 2018 BMJ open (10) | Prospective multicenter Spanish Cohort admission ICU (3) 2009-2012 | Spontaneous ICH | Mortality at 1 month | 336 Mortality 176 (52%) | - Age Median 62 IQR 50-70 - Female NA | No | AUC ROC 0.74 (0.69-0.79) | No | - HL GOF test p<0.001 - GiViTI calibration belt p<0.001 | No |
| | Schmidt et al 2018 Neurology (11) | Prospective cohort Chicago US (1) 2010-2017 | Spontaneous ICH | Unfav outcome (mRS 4-6) at 3 months | 372 Mortality at 3m: 153 (41%) Unfav outcome at 3m: 236 (63%) | - Age Mean 67 SD 14 - Female 51% | Likelihood ratio $X^2$ p<0.001 | AUC ROC 3m mortality 0.83 (0.79-0.88) 3m unfav outcome 0.85 (0.81-0.89) | No | - Histogram score vs observed mortality and poor outcome - Unweighted sum of squared errors test for GOF p>0.3 | No |
| | Sembill et al 2017 Neurology (12) | Prospective cohort Germany (1) 2007-2011 | Spontaneous ICH maximally treated | Mortality at 3 and 12 months Unfav outcome (mRS 4-6) at 3 and 12 months | 471 12-m mortality 142 (30.1%) 12-m unfav outcome 214 (45.4%) | - Age Mean 70 SD 12 - Female 45.4% | No | AUC ROC 12m mortality 0.69 (0.64-0.74) 12m unfav outcome 0.72 (0.67-0.76) | 3m mortality: PPV 37% 12m unfav outcome: PPV 74.7% Cutoff Youden | Histogram ICH score vs observed mortality | No |
| MICH | Sembill et al 2017 Neurology (12) | Prospective cohort Germany (1) 2007-2011 | Spontaneous ICH maximally treated | Mortality at 3 and 12 months Unfav outcome (mRS 4-6) at 3 and 12 months | 471 12-m mortality 142 (30.1%) 12-m unfav outcome 214 (45.4%) | - Age Mean 70 SD 12 - Female 45.4% | No | AUC ROC 12m mortality 0.65 12m unfav outcome 0.69 | No | No | No |
| ICH-GS | Chuang et al. 2009 Int J Qual Health Care (3) | Retrospective cohort Taiwan (1) 2006-2007 | Spontaneous ICH | Mortality at 1 month | 293 Mortality 40 (14%) | - Age Mean 60.6 SD 16.7 - Female: 30% | Accuracy 78.8% | AUC ROC 0.74 (0.65-0.83) | Cut off NA (highest Youden) Se, Sp, PPV, NPV, LR+, LR- (Table 4) | HL GOF test p>0,05 | No |
| | Naval et al. 2009 Neurol Res (8) | Registry of NICU USA (1) 1999-2006 | Supra-tentorial, spontaneous ICH Excl of prior mRS 2-5 | Mortality at 1 month | 125 mortality 29 (23%) | - Age 63.5 range 34-90 - Female 42% | No | No | Cut off ≥ 8 (Proba 50/50) PPV: 62.8% NPV 97.6% Se 93.1% Sp 83.3% | chi square test obs/pred overestimation of mortality of 11.2% (p=0,03) | No |
| Max ICH score | Schmidt et al 2018 Neurology (11) | Prospective cohort Chicago US (1) 2010-2017 | Spontaneous ICH | Unfav outcome (mRS 4-6) at 3 months | 372 Mortality at 3m: 153 (41%) Unfav outcome at 3m: 236 (63%) | - Age Mean 67 SD 14 - Female 51% | Likelihood ratio $X^2$ p<0.001 | AUC ROC 3m mortality 0.82 (0.78-0.86) 3m unfav outcome 0.88 (0.85-0.92) | No | - Histogram score vs observed mortality and poor outcome - Unweighted sum of squared errors test for GOF p>0.3 | No |

| Score | External validation | Study design Country (nb of centre) Date of incl. | Type of injuries | Outcomes | Sample size | Description of the population | Reporting of global performances | Reporting of discrimination | Reporting of classification | Reporting of calibration | Recalibration or updating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| VASOGRADE | Dengler et al. 2017 Eur J Neurol (14) | Hosp registry, managed in ICU Germany (1) 2009-2015 | aneurysmal SAH | - unfav outcome (mRS 3-6) at 12 months - unfav outcome (mRS 4-6) at 12 months | 423 208 (53.1%) | - Age mean 54.2 SD 13.7 - Female 69% | No | AUC ROC mRS 3-6 0.711 mRS 4-6 0.709 | No | Histogramme | No |
| SAH sum score / Hijdra score | Claassen et al. 2004 Crit Care Med (15) | Prospective cohort 1996-2002 USA (1) | SAH Excl AVM | Unfav outcome (mRS 4–6) at 3 months | 413 Unfav outcome 40.4% | - Age mean 54 SD 14- Female 71% | No | AUC ROC0.67 (0.61-0.73) | No | No | No |
| SAHIT | Mascitelli et al 2018 Neurosurgey (16) | Trial cohort admitted in ICU USA (1) 2003-2007 | aneurysmal SAH Excl lost to follow up (67) | - Unfav outcome (mRS 3-6) at 6 months - Mortality at 6 months | 338 Mortality: 38 (10,1%) Unfav outcome: 100 (29,6%) | - Age mean 54 SD 12 - Female: NA | $R^2$ Brier score Brier scaled (Figure 2) | AUC ROC Unfav outcome: core: 72.8 (66.8-78.9) Neuroimaging: 73.2 (67.1-79.2) Full: 73.4 (67.5-79.4) Mortality: core: 72.1 (62.1-82.2) neuroimaging: 73.9 (64.4-83.5) Full: 74.4 (65.1-83.8) | No | calibration plot Intercept Slope (figure 2) | No |
| HAIR / SAH score | Witsch et al. 2016 Ann Neurol (17) | - Prospective cohort (SHOP) 1996-2014 USA (1) - CONSCIOUS-1 Israel, Europe, North America (52) | SAH Excl AVM Excl missing predictors (97) multiple Imputation for mRS (351) by MCMCM | Unfav outcome (mRS 4-6) at 12 months | - SHOP: 1526 Unfav outcome: 1200 (79%) - CONSCIOUS: 413 | - SHOP: - Age mean 55.3 SD 14,5 - Female 68% -CONSCIOUS: median 55 | Nagelkerke $R^2$ Cox/Snell $R^2$ - COUNSCIOUS-1: N-$R^2$ 0.17 C/S $R^2$ 0.11 - SHOP: N-$R^2$ 0.45 C/S $R^2$ 0.32 | AUC ROC - CONSCIOUS-1: 71.8 (66.0-77.5) - SHOP: 88.3 (86.4-90.2) | No | No | No |
| | Dengler et al. 2017 Eur J Neurol (14) | Hosp registry, managed in ICU Germany (1) 2009-2015 | aneurysmal SAH | - unfav outcome (mRS 3-6) at 12 months - unfav outcome (mRS 4-6) at 12 months | 423 208 (53.1%) | - Age mean 54.2 SD 13.7 - Female 69% | No | AUC ROC mRS 3-6 0.739 mRS 4-6 0.737 | No | Histogram | No |
| | Abulhasan et al. 2017 Neurocrit Care (18) | Retrospective cohort in NICU Canada (1) 2010-2016 | SAH H&H 1-5 (multiple imputation MICE) | Mortality at discharge | 434 Mortality 14.10% | - Age 56 48-65 - Female 63.6% | No | AUC ROC 0.89 | No | Calibration curve intercept=-0.05 slope=0.77 | No |

| Score | External validation | Study design Country (nb of centre) Date of incl. | Type of injuries | Outcomes | Sample size | Description of the population | Reporting of global performances | Reporting of discrimination | Reporting of classification | Reporting of calibration | Recalibration or updating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| ISAT | Dijkland et al. 2016 Crit Care Med (13) | Hosp registry, managed in ICU The Nederlands (1) 2007-2011 (same biostat. as initial publication) | Presumed aneurysmal SAH Excl lost to follow up | Mortality at 2 months | 307 Mortality 94 (30.6%) | - Age Median 56 IQR 47–66 - Female 65% | No | AUC ROC WFNS at time of treatment: 0.89 WFNS at admission: 0.82 | No | - Plot - intercept and slope: slopes adm WFNS 1.417 ttt WFNS 1.959 intercept WFNS adm 1.502 WFNS ttt 2.248 | No |
| APACHE II | Claassen et al. 2004 Crit Care Med (15) | Prospective cohort 1996-2002 USA (1) | SAH Excl AVM | Unfav outcome (mRS 4–6) at 3 months | 413 Unfav outcome 40.4% | - Age Mean 54 SD 14 - Female 71% | No | AUC ROC 0.66 (0.60–0.73) | No | No | No |
| | Fallenius et al. 2017 Scand J Trauma Resusc Emerg Med (19) | Finnish Intensive Care Consortium (21) 2003 - 2012 | ICH [Excl missing data on predictors or outcome (n=1479)] | Mortality at 6 months | 3218 [1589 for recalib. to predict the mortality at 6m - 1629 for validation] Mortality: 1527 (48%) [Validation: 786 (48%)] | - Age: median 60 IQR 52-69 - Female: NA | No | AUC ROC 0.83 (0.81 - 0.85) | No | pvalue HL <0.001, pvalue GiViTI <0.001 | Yes To predict outcome at 6 months |
| | Huang et al. 2012 Eur J Neurol (6) | Registry of NICU Chine (1) 2000-2011 | primary pontine hemorrhage (ICH) | Mortality at 1 month | 75 mortality 31 (41%) | - Age 54.8 SD 12.7 - Female 21% | No | AUC ROC 0.919 (0.843 – 0.995) | Youden Cut off 16.5 Se 91.2% Sp 86.5% | - HL GOF test p=0.428 - Table obs / expected | No |
| | Moon et al. 2015 J Clin Neurosci (20) | Prospective cohort adm in ICU South Korea (1) 2001-2012 | ICH (60%) and ischemic strokes (40%) Excl 44 missing data | Mortality at discharge | ICH only: 300 Mortality 81 (27%) | - Age Mean 57.3 SD 17.2 - Female 48% | No | ICH only: AUC ROC 0.805 | No | ICH only: - Calibration curve - HL GOF test p=0.782 | No |
| | Rodriguez-Fernandez et al. 2018 BMJ Open (10) | Prospective multicenter Spanish Cohort (3) 2009-2012 | Spontaneous ICH | Mortality at discharge | 336 Mortality 181 (54%) | - Age Median 62 IQR 50-70 - Female NA | No | AUC ROC 0.80 (0.74-0.84) | No | - HL GOF test p=ns - GiViTI calibration belt p=0.43 | No |

| Score | External validation | Study design Country (nb of centre) Date of incl. | Type of injuries | Outcomes | Sample size | Description of the population | Reporting of global performances | Reporting of discrimination | Reporting of classification | Reporting of calibration | Recalibration or updating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SAPS | Handshu et al. 2005 J Neurol (22) | Handshu et al. Prospective cohort adm in ICU Germany (2) Period NA | ICH (54%) and ischemic stroke (46%) requiring endotracheal intubation | Mortality at 10 days, 3 months and 12 months | 90<br>3-mM 58.9%<br>12-mM 67.8% | - Age mean 64.3 SD 10.4<br>- Female 50% | No | AUC ROC<br>10 days:<br>0.67 (0.55-0.80)<br>3months:<br>0.75 (0.65-0.86)<br>12 months:<br>0.77 (0.67-0.88) | Youden<br>10d: cut off > 12<br>Se 66.5%<br>Sp 72.1%<br>3m: cut off > 12<br>Se 56.6%<br>Sp 83.8%<br>12m: cut off > 10<br>Se 77.0%<br>Sp 72.4% | No | No |
| SAPS II | Schuiling et al. 2005 Neurosurgery (23) | Retrospective cohort / The Nederlands (1) 2000 - 2002 | SAH with long lasting and severe vasospasm | Unfav outcome (mRS 4-6) at 3 months | 136<br>Unfav outcome:<br>65 (48%) | - Age<br>mean 55<br>range 17-93<br>- Female 73% | No | AUC ROC<br>0.85 (0.78-0.91) | No | No | No |
| | Schuiling et al. 2005 Neurosurgery (23) | Prospective cohort adm in ICU Germany (2) Period NA | ICH (54%) and ischemic stroke (46%) requiring endotracheal intubation | Mortality at 10 days, 3 months and 12 months | 90<br>3-mM 58.9%<br>12-mM 67.8% | - Age mean 64.3 SD 10.4<br>- Female 50% | No | AUC ROC<br>10 days:<br>0.68 (0.57 – 0.80)<br>3months:<br>0.77 (0.67 – 0.97)<br>12 months:<br>0.77 (0.66 – 0.88) | Youden<br>10d: cut off > 40<br>Se 75.9%<br>Sp 55.7%<br>3m: cut off > 36<br>Se 84.9%<br>Sp 62.2%<br>12m: cut off > 40<br>Se 72.1%<br>Sp 82.8% | No | No |
| | Celi et al. 2012 J Pers Med (24) | Retrospective analysis on prospective cohort USA (1) 1995 - 2006 | SAH | Mortality at discharge | MIMIC database<br>150<br>Mortality:<br>57 (25.6%) | NA | No | AUC ROC<br>0.84 | No | HL GOF test p<0.001 | No |
| | Czorlich et al. 2015 Acta Neurochir (Wien) (25) | ICU registry Germany (1) 2010 - 2014 | aneurysmal SAH | Unfav outcome (GOS 1-3) at 1 month | 263<br>Mortality:<br>49 (18.3%) | - Age<br>mean 54,4<br>SD 13,74<br>- Female 70% | No | AUC ROC<br>0,834 (0,771-0,896) | No | No | No |
| | Moon et al. 2015 J Clin Neurosci (20) | Prospective cohort adm in ICU / South Korea (1) 2001-2012 | ICH (60%) and ischemic strokes (40%) [Excl 44 missing data] | Mortality at discharge | ICH only:<br>300<br>Mortality 81 (27%) | - Age<br>Mean 57.3<br>SD 17.2<br>- Female 48% | No | ICH only:<br>AUC ROC<br>0.783 | No | ICH only:<br>- Calibration curve<br>- HL GOF test p=0.485 | No |
| | Huang et al. 2012 Eur J Neurol (6) | Registry of NICU Chine (1) 2000 - 2011 | primary pontine hemorrhage (ICH) | Mortality at 1 month | 75<br>mortality<br>31 (41%) | - Age 54.8<br>SD 12.7<br>- Female 21% | No | AUC ROC<br>0.890 (0.817 - 0.943) | Cut off 32.5 (Youden)<br>Se 82.4%<br>Sp 86.5% | - HL GOF test p=0.682<br>- Table obs / expected | No |

| Score | External validation | Study design Country (nb of centre) Date of incl. | Type of injuries | Outcomes | Sample size | Description of the population | Reporting of global performances | Reporting of discrimination | Reporting of classification | Reporting of calibration | Recalibration or updating |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Fallenius et al. 2017 Scand J Trauma Resusc Emerg Med (19) | Finnish Intensive Care Consortium (21) 2003 - 2012 | ICH [Excl missing data on predictors or outcome (n=1479)] | Mortality at 6 months | 3218 [1589 for recalib. to predict the mortality at 6m - 1629 for validation] Mortality: 1527 (48%) [Validation: 786 (48%)] | - Age: median 60 IQR 52-69 - Female: NA | No | AUC ROC 0.84 (0.82 - 0.86) | No | pvalue HL = 0.058, pvalue GiViTI = 0.014 | Yes To predict outcome at 6 months |
| | Barbieri et al. 2009 J Eval Clin Pract (2) | Registry of ICU Italy (1) Period NA | spontaneous intra parenchymal hemorrhage (ICH) | Mortality at 1 month | 81 mortality 49 (60.5%) | - Age 64.6 SD 13.8 - Female: 36% | No | AUC ROC 0.510 (0.377 - 0.642) | No | No | No |
| SAPS III | Basile-Filho et al. 2018 Medicine (21) | ICU registry Brasil (1) 2011-2016 | SAH | Overall mortality - Unknown horizon | 51 Mortality 14 (27%)? 37.8%? | - Age Mean 54 SD 10 - Female 67% | No | AUC ROC 0.73 (0.59-0.85) | No | No | No |
| SOFA | Fallenius et al. 2017 Scand J Trauma Resusc Emerg Med (19) | Finnish Intensive Care Consortium (21) 2003 - 2012 | ICH [Excl missing data on predictors or outcome (n=1479)] | Mortality at 6 months | 3218 [1589 for recalib. to predict the mortality at 6m - 1629 for validation] Mortality: 1527 (48%) [Validation: 786 (48%)] | - Age: median 60 IQR 52-69 - Female: NA | No | AUC ROC 0.73 (0.71 - 0.76) | No | pvalue HL <0.001, pvalue GiViTI <0.001 | Yes, To predict outcome at 6 months |
| | Basile-Filho et al. 2018 Medicine (21) | ICU registry Brasil (1) 2011-2016 | SAH | Overall mortality - Unknown horizon | Mortality 14 (27%)? 37.8%? | - Age Mean 54 SD 10 - Female 67% | No | AUC ROC Day1 0.62 (0.48-0.75) Day 3 0.77 (0.63-0.87) | No | No | No |
| SIRS summary score | Claassen et al. 2004 Crit Care Med (15) | Prospective cohort 1996-2002 USA (1) | SAH Excl AVM | Unfav outcome (mRS 4–6) at 3 months | 413 Unfav outcome 40.4% | - Age Mean 54 SD 14 - Female 71% | No | AUC ROC 0.57 (0.51–0.064) | No | No | No |

## References of the S4 Table


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